for eric 7/26/22

dan—
standards

Eric's Comments in Red 5/13/22

- To support the stability of larger palms, consider backfilling with washed builder's sand to pack more easily and uniformly.
- There is no benefit to amending the backfill with organic matter. Use organic matter as mulch several inches deep and several feet out from the palm's base. Tamp out air pockets.
- If stabilization is required for large palms:
  - Use 2 x 4 or 4 x 4 wooden bracing attached against one-foot lengths of 2 x 4 vertically strapped or banded around the trunk.
  - Protect the trunk with nylon, burlap, or other suitable material where the one foot lengths of 2 x 4 are secured.
  - Do not nail into the trunk; nailing will cause permanent wounds, and disease and insect entry sites.
  - Palms may also be secured with guy wires or cable instead of wooden bracing.
  - Do not stabilize palms by planting them deeper in the hole; although some palms survive deep planting, most do not.
- Construct an irrigation berm four to six inches around the root ball and hole.
- Provide a two- to three- inch layer of mulch around the base of the palm to encourage new root growth, conserve moisture, and suppress weeds.
- Irrigate deeply and thoroughly.

Post-planting care

- Schedule irrigations based on need, not a clock or calendar.
- Irrigate sensibly, keeping the root ball and backfill evenly moist but not saturated.
- Keep turf grass and weeds away from the trunk base.
- Maintain a regular and complete fertilizer program once the palm is fully established.

Division 33 - Utilities
33 10 00 - Water Utilities — add in a leak testing spec (Jason will locate one or use frost) or as noted elsewhere in specs

All underground plumbing utility piping shall have a minimum of 12" sand encasement. (or encasement may be needed)

All underground plumbing utility piping shall have a minimum depth of 24" — 36" unless otherwise approved by the administrative authority after all existing utilities have been potholed by the contractor.

All underground storm drainage and sanitary sewage piping shall be video-recorded prior to connection to existing systems. These videos shall be presented to campus building inspectors and representatives of the Plumbing Shop. Video footage to be furnish on CD or in an electronic format available for viewing. — remove from water add to both SS + SD specs

33 11 13 - Public Water Utility Distribution Piping
All underground domestic water piping 3" or less shall be schedule 80 PVC or HDPE SDR 11.

All underground domestic water piping 4" or larger shall be PVC C900.
All valves for underground domestic water 2" or larger shall be epoxy coated resilient wedge gate valves with 2" operating ~~head.~~ ~~nut~~ nut.

Provide Tracer wire For All Non metallic Pipe. See notes on PS 127 PS 138

Preferred Manufactures: Mueller Co. or Clow Valve. or equal

Valve well sleeve shall be 8" diameter SDR 35 pipe unless otherwise authorized by the University.

* ask dennis + chad on meters

I have an addition to 33 12 13 13
All backflow preventers will be tested and certified by a third-party licensed tester prior to being put into service preferably prior to being connected to the building service, fire system or irrigation. Retested and certified annually by a third-party licensed tester as the project is still active to stay in compliance. Then retested at project completion by a third party licensed tester .
Forms will be provide by the Cal Poly Plumbing Shop and A copy will be returned to the Cal Poly Plumbing Shop each time the device is tested.

Valve sleeve shall come up at minimum half the depth of the G5 box.

33 12 13.13 - Water Supply Backflow Preventer Assemblies
Preferred manufacturers are Zurn and Wilkins~~, and Febco.~~

All backflow preventers shall meet AWWA standards.

All backflow preventers installed by the contractor will be ~~rested~~ tested by a certified tester and documentation will be turned over to a campus representative before lines are put into service.

33 12 19 - Water Utility Distribution Fire Hydrants
Fire hydrant spacing shall comply with CFC and City of San Luis Obispo Fire Marshall's requirements. ? and/or DSFM

All site building will be fully protected by fire sprinklers so a 75% reduction in fire flow can be applied with the authorization of the City San Luis Obispo's Fire Marshall.

Reduced pressure principle back flow preventers (RP devices) are required at any connection that would potentially contaminate the water supply. The RP device shall be a double detector check valve assembly at the connections to the fire risers and stand pipes.

All water pipes shall be installed with a minimum of 36" of cover.

Water meters shall be installed at every water service connection per the Plumbing Narrative.

Valves shall be arranged so that the system can be operated in a manner where no more than 5 fire protection apparatuses (hydrants, risers, stand pies, etc.) will be removed from service if repairs to the system are necessary.

Vertical clearance of all water pipes shall comply with State Department of Health separation requirements and water shall always be above the sanitary sewer.

Horizontal clearance shall comply with State Department of Health separation requirements with a minimum 10' separation of sewer and water measured from outside of pipe.

Fire hydrants shall be placed at a maximum of 300' from all portions of the buildings per City of San Luis Obispo Fire Marshall's requirements.

Fire department connections shall be placed a maximum of 40' from a fire hydrant per City of San Luis Obispo Fire Marshall's requirements.

Provide a minimum residual pressure at the roof elevation of 20 psi at fire flow (1,500 gpm) + average day demand (ADD).

Pipes less than 4" diameter shall be schedule 80 PVC or SDR11 HDPE.

Pipes from 4" to 12" in diameter shall be C900 DR18 PVC.

Provide Tracer wire For All Non metallic Pipe

All fittings shall be ductile iron.

See notes on Pg 127 Pg 138

TRACER WIRE

· See comments on pg. 138 of 143 in SS section.
· Wires shall terminate at valve box of daylight pipe location accessible to staff.

All gate valves shall be non-rising stem with resilient wedge seats.

2" operating nut is preferred on all underground gate valves.

All fire hydrants shall comply with City of San Luis Obispo Public Works Engineering Standards. requirements and have solid bronze bodies, shall be wet barrel, with one 2-1/2", and two 4-1/2" connections.

within 20' per NFPA

~~Backflow~~RP devices for fire connection shall be double detector check valves, with lockable OS&Y stems, leak detection meters, and shall employ reduced pressure principle check valves.

2nd valve cut in

Tee is the preferred valve method for hydrants. (cut in tee with valve)

33 31 00 - Sanitary Utility Sewerage Piping

All sanitary sewer lines 4" or larger shall be SDR-35 PVC, lines 3" and smaller shall be DWV PVC or cast iron.

Provide tracer wire w/ All nonmetallic pipe

CP standard.

All sanitary sewer manholes shall meet City of San Luis Obispo Public Works Engineering Standards. Transition from existing clay and cast iron pipe to PVC shall be made with ~~Fernco~~ Rubber coupling. With steel band (strongback) or equal. Strong back

Provide documents or certified test results indicating the pipe furnished meets all specified requirements. Satisfactory documents include pipe manufacturer certificate indicating that the pipe has been:

1. Sampled
2. Tested
3. Inspected

In compliance with the ASTM specifications.

36" (or encasement may be needed)

Sanitary Sewer Design Criteria

All sanitary sewer laterals and mains shall be installed with a minimum of 48" 24" of cover.

Curved sewer mains shall be installed with a minimum curvature in compliance with manufacturer's recommendations. Joint deflections are not permissible.

All sanitary sewer mains shall have a minimum 8" diameter.

Sewer mains shall be designed with a minimum 2 fps and a maximum 10 fps half-full flow velocity and the following minimum slopes:

- 8" 0.50%
- 10" 0.35%

Force mains shall be designed with a minimum velocity of 4 fps and a maximum velocity of 8 fps.

Sewer laterals shall be 4" minimum diameter with a minimum slope of 2%.

Sewer mains shall have manholes at 300' maximum spacing, at angle points, at the beginning or ending of curves, clean outs may be used at the beginning of runs less than 150' long, otherwise a manhole is required.

Include reference - Double clean out design should be included (threaded plugs

Service laterals shall connect to the main with a wye pipe fitting and a clean out at the building connections and angle points. Laterals longer than 100' long shall require an additional clean out at approximately mid-point or at 100' maximum spacing. See campus stds. for cleanouts.

Sewer pipes shall be installed a minimum of 12" below water lines in compliance with State Department of Health separation requirements.

Horizontal clearance between sewer and potable water mains shall be a minimum of 10' from outside of pipe in compliance with State Department of Health separation requirements.

Lift Station Design Criteria

Pumps shall be an alternating duplex system with Individual pumps sized for PDD (250 gpm); Total dynamic head = ±120'. A triplex pumping system may be installed at the Design Builder's discretion and with University approval.

Accessories shall include an alarm system tied into a central monitoring station and odor control.

A stand by generator shall be included in the project to provide emergency power for the lift station. The generator must be sized to power both pumps in a duplex system or at least two pumps in a triplex system.

Sanitary Sewer Materials

New sewer Forcemain pipeline material must be fused HDPE. When repairing existing pipeline, comply with all sewer repair sections.

All gravity mains and laterals shall be SDR 35 PVC.

All force mains shall be Schedule 80 PVC or C900 DR 14 PVC.

Provide Tracer wire with All non metallic Pipe See note on Pg 127

All Manholes shall be pre-cast concrete with eccentric cones, and H20 traffic rated covers and frames.

Lift station wet wells shall be precast or cast in place concrete lined with a minimum 125 mils of type B polyurethane coating. Manhole walls and cone shall be coated. Do not coat step or channel at the bottom, which prevents a slippery surface.

Lift station pumps shall be submersible pumps with oil filled, explosion proof motors with moisture sensors. Pumps shall be mounted to a factory provided rail system.

spray on coating

***High Density Polyethylene* Pipe**

Use:

1. Virgin grade
2. ~~High molecular weight~~
3. Standard Dimension Ration (SDR) 17
4. Iron Pipe (IPS)

High Density Polyethylene (HDPE) pipe made in diameter and tolerances in compliance with the latest version of ASTM D3035.

Furnish complete with all fabricated fittings, and other appurtenances as necessary, for a complete and functional system.

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The pipe must be free of:

1. Visible cracks
2. Holes
3. Foreign inclusions, or
4. Other defects

Any pipe not meeting these criteria will be rejected.

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The pipe must be clearly marked with the following:

1. Name and trademark of manufacturer
2. Nominal pipe size
3. Dimension ration
4. The letters PE followed by the polyethylene grade per the latest version ASTM D1248
5. Hydrostatic design basis in psi
6. Manufacturing standard reference
7. A production code from which the date and place of manufacture can be determined.

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The material must be listed by the Plastic Pipe Institute (PPI) with a designation PE 3408 and have a minimum cell classification of:

1. 345434C
2. D, or
3. E

As described in latest version of ASTM D3350.

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Pipe material must meet the requirements for:

1. Type III
2. Class B or C
3. Category 5
4. Grade P34

Material as described in latest version of ASTM D1248

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Provide pipe with interior wall color of:

1. White
2. Gray or
3. Light green

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Provide pipe with exterior wall color of:

1. Black
2. Gray or
3. Light green

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Provide submittals on furnished pipe from manufacturer certifying pipe is in compliance with:

1. Specifications
2. Codes
3. Standards

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Any pipe segment that has cut in the pipe wall exceeding 10 percent of the wall thickness must be cut out and removed from the site.

Store pipe so that it is not deformed. or subject to UV exposure

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Polyvinyl Chloride (PVC) Pipe
Furnish pipe in 20-foot lengths with integral wall belled ends and elastomeric joint and solid wall. Pipe and fittings must be free of imperfections and clearly marked with name of manufacturer.

Minimum pipe stiffness (F/y) at 5 percent deflection is 46 psi for all sizes when calculated in compliance with ASTM Designation D 2412.

Pipe must have minimum Standard Dimension Ration (SDR) of 35 and pipe stiffness of 46 psi.

Pipe color must be green.

PVC Pipe 4 to 15 Inch Diameter
PVC Pipe must conform to the requirement of latest version of ASTM specification D 3034.

PVC Pipe 18 to 27 Inch Diameter
PVC Pipe must conform to the requirement of latest version of ASTM Standard Specifications F 679.

PVC Pipe 30 to 48 Inch Diameter
PVC Pipe must conform to the requirement of latest version of ASTM Standard Specifications F 794.

Ductile Iron Pipe
Ductile iron pipe must be:

1. Centrifugally cast.
2. Ductile iron pipe.
3. Gasketed push on joints appropriate for use in a wastewater environment such as Polychloroprene, Ethylene Propylene Diene Monomer, or an approved equal.
4. A pressure class 150 for pipe with 3 feet or less of cover.
5. A pressure class of 350 for pipes with 3 feet or less of cover or exposed above grade.
6. Coated on exterior.
7. Lined with fusion bonded epoxy, polyurethane or approved equal.

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Ductile iron pipe must be encase in polyethylene casing material. Casing material must be:

1. Tube type
2. Conform to the latest ANSI/AWWA C105 Standard.

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Polyethylene casing must extend over:

1. Tees
2. Bends
3. Couplers at the end of a Section of ductile iron where it connects to a different type of pipe
4. Close casing at the end (dead end) of a pipe

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Exposure to air and sunlight must be kept to minimum for either type "A" or type "C" encasement material.

Sewer Lateral Pipe
New and repaired sewer lateral pipe may be:

1. PVC SDR 35
2. PVC Schedule 40
3. HDPE SDR 17
4. ~~ABS Schedule 40~~

I don't believe this ok by P.S.

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Pipe joints must be gled or fused.

Joints and Fittings

HDPE

HDPE Pipe and fittings must be in compliance with the latest version of:

1. ASTM F714
2. ASTM D3261

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Joints and Fittings for HDPE must be of the same manufacturer as the pipe and the same SDR rating.

PVC

PVC pipe must have a rubber ring bell and spigot joints providing a water tight seal and allowing for contraction and expansion. The bell must consist of an integral wall Section stiffened with two PCV retainer rings that securely lock the solid cross Section rubber ring into position.

All fittings and accessories must be as manufactured and furnished by the pipe supplier, or approved equal, and have bell and/or spigot configurations identical to that of the pipe. All fittings must be of the same material as the pipe and same SDR rating, unless specified otherwise.

Ductile Iron

Use restrained fittings for exposed ductile iron pipe, such as bridge crossings. Restrained fittings must be Flex-Ring by American Ductile Iron, TY FLEX by U.S. Pipe, or approved equal which use a factory weld as part of the restraining system.

Repair Joint

Use strong back RC couplings or equal meeting the following requirements:

1. ~~Flexible sewer couplings and transition couplings~~ delete. not strong back
2. Comprised of an elastomeric sealing component
3. Type 316 series stainless steel tension components (end clamps and shear rings).
4. Shear rings must have am minimum thickness of 0.012 inches
5. End clamps must have "bolts" as their means of tightening (not worm gears).

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Couplings must be appropriately sized for the pipe materials being joined, without the need for bushings.

HDPE Pipe with fused ends must be repaired with HDPE pipe with gused joints. Strong back couplings must not be used.

Sewer Lateral Joints (New and Replacement)

Sewer lateral pipe must be joined using glued joints and fittings or fused.

Concrete

Use class 2 concrete.

Use minor concrete for:

1. Manholes
2. Pipe junctions
3. Jacketing

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Use fifteen percent approved pozzolan replacement for manhole construction.

Precast concrete manhole Sections must comply with the most current version of ASTM specification C-478-61T or AASHTO-M170.

All manholes must be watertight, and the floor sloped for a smooth monolithic trowel finish. The interior finish of the manholes must be smooth. a motar surface.

Mortar

To make mortar, use one part of Type II Portland cement and two parts sharped grained particles that are:

1. Clean
2. Hard
3. All passing a # 4 sieve

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Mix mortar in a machine or water tight box. Accurately measure and thoroughly mix mortar to a uniform consistency. Use mortar immediacy after mixing. Do not remix mortar that begins to harden prior to placement.

Pipe Installation

Sanitary sewer lines must be water tight. Install pipe to ensure the system is water tight throughlout the component parts, particularly at the pipe joint.

Do not:

1. Cut
2. Gouge
3. Score or
4. Damage pipes

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When:

1. Unloading
2. Handling
3. Storing
4. Installing

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Pipe Laying

Lay the pipe in perfect conformity to the design line and grade obtained for each pipe by measuring down from a tightly stretched line running parallel with the grade.

Lay all pipes continuously uphill.

Install pipe and fittings for underground gravity sewers in compliance with the latest version of ASTM Standard D-2321. Lay bell and spigot pipe, with the bell of the pipe upgrade.

Pipe Bursting and Reaming

Install sewer pipe by pneumatic pipe bursting or pipe reaming. Install pipe in compliance with the pipe manufacturer's recommendations. For pipe busting installation, use pneumatically operated equipment with a pipe bursting head attached to HDPE pipe.

Locate, expose, disconnect and isolate existing sewer laterals from sewer main before pipe installation work begins. When pipe reaming, you must prevent drilling fluid from entering sewer laterals.

Submit to the Engineer for review and approval a sewer installation plan which includes insertion and reception pit location.

For pipe bursting work, use a constant tension pneumatic tool used in conjunction with a constant tension hydraulic winch. Size the winch based on the diameter and the depth of the pipe to be replaced. The constant tension winch must be sufficient sized to pull one continuous length of pipe between approved winching points.

The void created by the device must be sufficient in size to accommodate the pipe which is installed immediately after the void is formed. The void must not be so large that pipe displacement or pavement settling occurs. Allow new sewer pipe to relax for twelve hours prior to final connection to manholes.

If you cannot complete pipe bursting or reaming without damage to existing closely places lines or pavement, you may request authorization from the Engineer to place new pipe with traditional open-cut trenching. If you encounter an obstruction that prevents the bursting or reaming tool from continuing, you must:

1. Stop the operation
2. Notify the Engineer
3. Excavate the obstruction
4. Remove the obstruction.

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Any pavement heaving, or utility damage caused by pipe bursting or reaming work must be repaired at no additional cost to the city or utility company.

or ~~Campus Inspector~~ University

If you use any material or method that is not approved by the Engineer, you must remove the work and replace as directed by the Engineer. or ~~Campus Inspector~~ university

If an obstruction is found during testing, remove the obstruction. Remove and replace Section of pipe is damaged.

HDPE Pipe Joint

Join HDPE pipe by:

1. Heat fusion welding
2. Electrofusion fitting or
3. Equal as approved by the Engineer.

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All connections to the sewer pipe must be water tight, flush with the edges of the sewer pipe with clean uniform cuts.

Perform heat fusion welding in compliance with the pipe manufactuer's recommendations and ASTM D2657. Fusion equipment used must be capable of meeting all conditions recommended by the pipe manufacturer including, but not limited to:

1. Fusion temperature
2. Alignment
3. Fusion pressure

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Fusion equipment must only be operated by technicians who have been certified by the pipe manufacturer or supplier. Document and furnish to the Engineer technicians certification in a submittal.

Use a fire-retardant bag or suitable enclosure for the heater plate to facilitate control of heating process and to protect the heater plate surfaces from dirt and other debris when not in use.

Clean heater plate surfaces regularly to prevent accumulation of fusion welding residues or other substances that may result in faulty pipe joining. The heater plate must be equipped with suitable means to measure the temperature of plate surfaces and to assure uniform heating such as thermometers or pyrometers.

Joint strength must be equal to that of the adjacent pipe. Clean the pipe ends with a cotton or non-synthetic cloth to remove:

1. Dirt
2. Water
3. Grease
4. Other foreign materials.

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Cut pipe ends square and carefully aligned just prior to heating.

After achieving the proper melt pattern, bring the pipe ends together in a firm, rapid motion applying sufficient pressure to form a pipe bead (1/8 to 3/16 inch in height) around and inside the entire circumference of the pipe. Remove pipe bead before welding the next joint of pipe.

Use tools designed for and approved by the manufacturer and supplier for joining pipe.

Sand Traps

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Furnish and install sand traps or other debris catching measure approved by the Engineer during the work. Debris catching devices must always be installed during construction. You assume all costs associated with any damage resulting from construction materials entering the wastewater system or treatment facility.

Bypass Pumping

Submit a bypass pumping plan for approval by Engineer at the pre-installation ~~pre-construction~~ meeting, or prior to work beginning. At a minimum the plan must include:

1. Pump size and type
2. Backup pump size and type
3. Contingency plan for pump failure to ensure continuous bypass operations.

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The bypass system must be free from leaks. The bypass pumping plan must address access to:

1. Driveways
2. Cross streets
3. Pedestrian crossings.

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Manholes

Construct manholes per Engineering Standards.

Sewer Laterals

Sewer laterals must be tied over as shown. Notify the Engineer or ~~Campus Inspector~~ University immediately upon discovering any lateral not shown, or any lateral that appears to be dry and out of service. The Engineer or ~~Campus Inspector~~ University will then determine if

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lateral is out of service and not needed. If the lateral is out of service and not needed, the Engineer will direct you to abandon the lateral. In these cases, the pay item for lateral tie over will not be reduced and will be paid to compensate you for all time, equipment, labor, materials for sewer lateral abandonment.

Existing Sewer

Existing Manholes
Existing manholes must be:
1. Adjusted to grade
2. Remodeled or
3. Abandoned

As shown and in compliance with Engineering Standards and Section 15.

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Existing manholes may have large cast in place concrete bases. No additional payment will be made for the removal of existing bases as needed to complete the work.

Oversize manholes may require a manufactured concrete reduction ring prior to setting the new manhole ring and cover.

Abandonment of Sewerlines
Sewer facilities taken out of service must be abandoned in compliance with engineering Standard 6050.

Provide the Engineer 48-hour notice prior to abandoning sewer laterals. Cut off the sewer lateral at the main and plug pipes with class 3 concrete a minimum of 12-inches into the pipe, away from the sewer main pipe. Provide a minimum five-foot by five-foot excavation with shoring at the sewer main, adequate for University to remove the existing wye and replace it with new section of pipe. Provide:
1. Backfill
2. Compaction
3. Surface restoration

For the excavation.

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Repair
Repair cut sewer facilities using new pipe of material the same diameter. If the existing sewer pipe material complies with materials, use that same pipe material.

Center a continuous Section of new pipe at the repair location. Repair must be water tight and placed at the same grade. Prior to backfilling excavation, place level on repaired portion of sewer, in the prewsence of the Engineer, to confirm line and grade. Backfill, compact and restore surface improvements.

Repair must be documented with:
1. Location
2. Repairs made
3. Photos
4. Guarantee letter
5. Interior video inspection of pipeline, when directed by the Engineer.

Provide hardcopy of all documents to owner. Provide electronically, all documents to the Engineer.

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Testing

or ~~Campus~~ Inspector ~~per AWWA Std.~~ University

Air Test

After the pipeline is in place and the joints made, you must air test the sewer in the presence of the Engineer. Air test procedure is as follows:

1. A maximum of 400 feet of sewer pipe will be tested at one time.
2. Plug and brace securely all outlets.
3. Introduce air into test Section until internal pressure is 4.0 psi. If sewer pipe is placed in ground water, calculate ground water pressure and add that additional pressure to internal pressure used for test.
4. Maintain an internal test pressure by adding air as need for a minimum time of 2 minutes.
5. Measure the time required for pressure to drop from 3.5 psi to 2.5 psi. Do not introduce new air into test Section during measurement.

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Minimum permissible pressure discharge time as follows in seconds
(time to drop pressure from 3.5 psi to 2.5 psi)

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Sewer Main		4-inch Sewer Lateral				
Diameter	Length	Sewer Lateral Length				
Inches	Feet	0 feet	100 feet	200 feet	300 feet	400 feet
6 & 8	0	0 seconds	20 seconds	40 seconds	50 seconds	70 seconds
	50	40 seconds	50 seconds	70 seconds	90 seconds	80 seconds
	100	70 seconds	90 seconds	100 seconds	100 seconds	90 seconds
	150	110 seconds	120 seconds	110 seconds	100 seconds	100 seconds
	200	140 seconds	120 seconds	110 seconds	110 seconds	100 seconds
	300	140 seconds	130 seconds	120 seconds	110 seconds	110 seconds
	400	140 seconds	130 seconds	120 seconds	120 seconds	110 seconds
10	50	50 seconds	70 seconds	90 seconds	100 seconds	90 seconds
	100	110 seconds	130 seconds	120 seconds	110 seconds	110 seconds
	200	170 seconds	150 seconds	140 seconds	130 seconds	120 seconds
	300	170 seconds	160 seconds	150 seconds	140 seconds	130 seconds
	400	170 seconds	160 seconds	150 seconds	150 seconds	140 seconds
12	50	80 seconds	100 seconds	110 seconds	110 seconds	110 seconds
	100	160 seconds	170 seconds	150 seconds	140 seconds	130 seconds
	200	200 seconds	180 seconds	170 seconds	160 seconds	150 seconds
	300	200 seconds	190 seconds	180 seconds	170 seconds	160 seconds
	400	200 seconds	190 seconds	180 seconds	180 seconds	170 seconds
15	50	120 seconds	140 seconds	160 seconds	140 seconds	130 seconds
	100	250 seconds	220 seconds	190 seconds	170 seconds	160 seconds
	200	260 seconds	230 seconds	220 seconds	200 seconds	190 seconds
	300	260 seconds	240 seconds	230 seconds	220 seconds	210 seconds
	400	260 seconds	240 seconds	230 seconds	220 seconds	220 seconds

Sewer Main		6-inch Sewer Lateral				
Diameter	Length	Sewer Lateral Length				
Inches	Feet	0 feet	100 feet	200 feet	300 feet	400 feet
6 & 8	0	0 seconds	40 seconds	80 seconds	100 seconds	100 seconds
	50	40 seconds	70 seconds	110 seconds	110 seconds	110 seconds
	100	70 seconds	110 seconds	120 seconds	110 seconds	110 seconds
	150	110 seconds	120 seconds	120 seconds	120 seconds	110 seconds
	200	140 seconds	130 seconds	120 seconds	120 seconds	120 seconds
	300	140 seconds	130 seconds	120 seconds	120 seconds	120 seconds
	400	140 seconds	130 seconds	130 seconds	120 seconds	120 seconds
10	50	50 seconds	90 seconds	120 seconds	120 seconds	110 seconds
	100	110 seconds	140 seconds	130 seconds	130 seconds	120 seconds
	200	170 seconds	150 seconds	140 seconds	140 seconds	130 seconds
	300	170 seconds	160 seconds	150 seconds	140 seconds	140 seconds
	400	170 seconds	160 seconds	150 seconds	150 seconds	140 seconds
12	50	80 seconds	120 seconds	140 seconds	130 seconds	120 seconds
	100	160 seconds	170 seconds	150 seconds	140 seconds	140 seconds
	200	200 seconds	180 seconds	170 seconds	160 seconds	150 seconds
	300	200 seconds	190 seconds	180 seconds	170 seconds	160 seconds
	400	200 seconds	190 seconds	180 seconds	180 seconds	170 seconds
15	50	120 seconds	160 seconds	160 seconds	150 seconds	140 seconds
	100	20 seconds	210 seconds	190 seconds	170 seconds	160 seconds
	200	260 seconds	230 seconds	210 seconds	200 seconds	190 seconds
	300	260 seconds	240 seconds	220 seconds	210 seconds	200 seconds
	400	260 seconds	240 seconds	230 seconds	220 seconds	210 seconds

PVC Joints
Joint tightness is measured by assembling two Sections of pipe in compliance with the manufacturer's recommendations.

Subject the joint to an internal hydrostatic pressure of 25 psi for one hour. Consider any leakage a failure of the test requirements.

Testing of Force Mains
Test force mains according to the following procedure:

Fill each section of pipe with water and expel all air. Allow pipe to set for a minimum of 24 hours. Refill pipe and pressure pipe to:

1. 150 psi, or
2. Service pressure plus an additional 50 psi

Whichever is greater. Maintain pressure for two hours. Replace any portion of line that fails and retest. Maximum allowable leakage is 4.17 gallons per hour per mile per nominal inch of diameter.

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Manhole Vacuum Testing
Vacuum test all newly constructed manholes prior to placing any backfill around manhole and again after manhole is raised to finish grade. Provide the Engineer 24-hour notice prior to each test.

You must prepare the manhole as follows:

1. Plug all inlets to the manhole
2. Place a test head in the top of the manhole
3. Inflate a seal

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Place a vacuum of 10 inches of mercury on the manhole and measure the time for the vacuum to drop to 9 inches of mercury. The manhole meets requirements if the measured time for the vacuum drop meets or exceeds the value from the following table:

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Manhole Depth	Manhole Diameter	
	4 feet	5 feet
4 feet	10 seconds	13 seconds
6 feet	15 seconds	20 seconds
8 feet	20 seconds	26 seconds
10 feet	25 seconds	33 seconds
12 feet	30 seconds	39 seconds
14 feet	35 seconds	46 seconds
16 feet	40 seconds	52 seconds
18 feet	45 seconds	59 seconds
20 feet	50 seconds	65 seconds

If the manhole fails the vacuum test, provide the necessary repairs to make the manhole pass the vacuum test. Retest accordingly.

TRACER WIRE

- Shall be #12 coated, stranded wire with water proof connections
- Shall be ran on all pipelines and manholes.
- Wire shall be ran and terminated at manhole ring, inside at the last grade ring.

✓

33 40 00 - Storm Drainage Utilities

Storm water management systems must meet RWQCB post-construction requirements as listed in the Post-Construction Storm water management Requirements for Development Project in the Central Coast Region (Resolution No. R3-2013-0032).

Project storm drain systems shall be designed for the 25-year design storm with 1' of freeboard at inlets, and overland relief shall be provided for the 100-year design storm.

Pipes less than ~~12"~~ 18" or less in diameter shall be SDR 35 PVC at a minimum of 0.5% slope; all pipe ~~12" or~~ 18" or greater shall be dual wall type S Corrugated Polyethylene Pipe (CPP) with water tight joints at a minimum of 0.5% slope. All pipes shall have a minimum cover of 24". All pipes within the street shall be a minimum of 18" diameter.
36" (or encasement may be needed)

Manholes or catch basins are required at angle points, at beginning or ending of curves, or on runs longer than 300'.

Provide minimum cover on all pipes in accordance with manufacturer's recommendations and not less than 12".

Provide a minimum of 12" of clearance to all other underground utilities.

Provide a minimum of 12" of cover from bio-retention swale gravel beds to top of pipes.

All manholes, catch basins, and junction boxes shall be concrete with cast iron lids or grates. All grates and lids shall be ADA compliant, bicycle friendly, and shall be traffic rated where there is a potential for vehicular traffic, including on sidewalks. Galvanized grates and lids are not allowed to prevent heavy metal contamination of storm water.

The bio-retention swale overflow inlets and drain pipe shall be designed to have adequate capacity to convey the 100-year storm with a minimum of 12" of free board in the swale.

Roof drains down spouts shall not directly connect to the storm drain system and the runoff should be directed to vegetated areas where possible. The intent of the storm drain system is to allow for bio-filtration of the storm water runoff and infiltration. Drainage should be conveyed in surface swale or by sheet flow where possible and pipes should be avoided unless necessary.

Storage can be provided in high porosity aggregate or perforated pipes or storm chambers surrounded by high porosity aggregate. Storage can also be provided in the aggregate under porous pavers. The system should be designed to infiltrate completely in 48 hours or less. Excess runoff to these systems shall be connected to the storm drain collection system.

Contractors are responsible for preparing a Storm Water Control Plan (SWCP) in accordance with Post-Construction Storm water management Requirements for Development Project in the Central Coast Region (Resolution No. R3-2013-0032), to be approved by the RWQCB.

Deflection
Following the:
1. Placement

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TRACER WIRE

-See comment on pg. 138 of 143 in SS Section.

2. Backfill
3. Compaction

Prior to permanent pavement, clean and measure pipe for obstruction such as:

1. Deflections
2. Joint offsets
3. Lateral pipe intrusions.

Allowable internal diameter is determined using appropriate size mandrel. Prior to use, the mandrel must be certified by the Engineer or by another entity approved by the Engineer. Use of an:

1. Uncertified mandrel or
2. An altered mandrel

Will invalidate test. If the mandrel fails to pass, the pipe will be deemed to be over deflected.

The mandrel must:

1. Be rigid
2. Be nonadjustable
3. Have an odd-numbering-legs (9 legs minimum)
4. Have an effective length not less than its nominal diameter
5. Be fabricated of steel or aluminum
6. Be fitted with pulling rings at each end
7. Be stamped or engraved indicating the:
8. Pip material specification
9. Nominal size
10. Mandrel outside diameter.

Copy this section to sewer spec and leave it here too

add into Sewer Spec

Using the manufacturer's specified internal diameter of pipe, maximum vertical deflection must not exceed:

1. 95 percent – for nominal diameter pipe less than or equal to 12 inches
2. 96 percent – for nominal diameter pipe less than or equal to 30 inches
3. 97 percent – for nominal diameter pipe greater than 40 inches

For pipes equal to or smaller than 24 inches in internal diameter, pull the mandrel through the pipe by hand. For pipes greater than 24 inches in internal diameter, deflections may be determined by mandrel or by a method submitted to and approved by the Engineer. If a mandrel is selected it must conform to the requirements in this section.

Any over deflected pipe must be uncovered to remove the compact soil loading. Once uncovered if the pipe can pass the mandrel it may remain. If not, remove and replace the damaged pipe. In all cases, the Engineer will determine whether the pipe may remain or must be replaced. Any pipe subjected to any method or process other than uncovering, even if successful to remove over deflection, must be removed and replaced with a new Section of pipe.

All costs incurred by ~~you~~ Contractor (typ) attributable to:

1. Mandrel testing
2. Deflection testing
3. Repairs
4. Any delays

Are borne by ~~you~~ Contractor at no cost to the University.

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Television Inspection (Video Inspection)

The University will video inspect (CCTV) all public sewer pipe systems prior to acceptance. Provide the Engineer three weeks' notice prior to placement of final paving or surface restoration. Allow one working day per 2,000 linear feet of sewer main to be video inspected. Installations which do not conform to the requirement must be reconstructed and re-video inspected at your expense.

If you are required to submit video inspection to the Engineer for review, furnish video on flash drive. or CD CCTV reports shall be National Association of Sewer Service Companies (NASSCO) Pipeline Assessment and Certification Program (PACP) certified with no modifications. Every section of sewer (manhole to manhole) shall be identified by audio and alphanumeric on the video display and shall include:

1. Project name
2. Municipality
3. Street name
4. University designated GIS manhole numbers
5. Sewer diameter and length
6. Date of inspection

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2dd Into Sewer spec

Video inspection recordings and reports must be completed for gravity conveyance systems per the following requirements:

1. The pipeline video inspection must be submitted on USB drive.
2. The inspection recording must be of adequate resolution to display pipe, potential pipe defects, and pipe joints.
3. Audio and written notes must be recorded in the video.
4. A 1-inch cylindrical gauge may be required in the video inspection to determine if the pipe segment has a grade deficiency.
5. General convention of the recording will travel downstream and must automatically track the pipeline length (in feet) from the start of the inspection to the end of the inspection.
6. Inspection report must include the project location, a scaled plan of the pipe segment(s), and a summary of inspection findings.

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Cleaning

After the final air test has been satisfactorily completed, sewer shall be cleaned using High-Velocity (Hydro-cleaning) equipment only. Cleaning shall be with clean water with a minimum 2,000 psi @ 50 gpm standard cleaning nozzle. Cleaning shall be performed starting at the furthest upstream segment (manhole to manhole) proceeding downstream. Each segment shall be cleaned from its downstream manhole. All debris shall be vacuumed and removed from each set-up location.

All foreign material must be removed from:

1. Pipes
2. Manholes
3. Cleanouts

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Prior to being placed into service. Remove all material from sand traps or debris catchers in manholes prior to removing the sand trap or debris catcher.

33 41 00 - Storm Utility Drainage Piping

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All storm sewer lines 4" or larger shall be SDR-35 PVC, lines 3" and smaller shall be DWV PVC or cast iron.

Back fill - river run sand

All manholes shall meet City of San Luis Obispo Public Works Engineering Standards.

Transition from existing clay and cast iron pipe to PVC shall be made with Ferno Rubber coupling.

33 51 00 - Natural-Gas Distribution

Cal Poly owns and maintains the underground natural gas distribution systems throughout the campus. Gas distribution pressure is 5-~~8~~10 PSI ~~and 20- 40 psi~~. The ~~campus~~ Southern California Gas Companies also has a 60 PSI distribution system.

not currently applicable

Commented [MDS6]: Albert to check.

Provide a gas regulator and meter at each new building.

For new buildings to be connected to the campus natural gas system, the anticipated additional gas demand should be identified early during preliminary planning. This anticipated gas demand should be submitted to the Principal Engineer for Cal Poly Physical Planning and Construction. Improvements to the campus system may be required to accommodate the additional demand. The Principal Engineer shall identify a suitable point of connection to the campus system and what system improvements may be necessary to accommodate the new building.

33 51 13 - Natural-Gas Piping

All underground gas piping shall be polyethylene SDR-11 yellow and installed by a contractor/employee with current certifications in the type of fusion they are using.

Provide Tracer Wire For All Non metallic Pipe See Notes Pg 127 Pg 138

All underground isolation valves for gas lines shall be polyethylene with 2"~~-ich~~ operating nut. accessible in a G-5 box.

Acceptable Piping Materials

- Above Grade 2" and smaller: Schedule 40 black steel pipe with treaded 150 pound black malleable iron fittings. Paint all exterior and exposed piping to prevent rust.
- Above Grade 2-1/2" and larger: Schedule 40 black steel pipe with welded steel fittings. Paint all exterior and exposed piping to prevent rust.

or mega press

- TRACER WIRE → see comments on 138 of 143 fuss section

Lb or mega press

Building Gas Pressure:

Regulate gas before entering buildings from 5~~10~~ PSI to 7 inches water column. Size the building gas distribution system based on a pressure of 7 inches water column. For some special high gas use applications such as large boilers, a 2 PSI mechanical room gas pressure may be allowed when approved by the University Representative and permitted by code.

Regulators:

Gas Regulators should be specified to include an under pressure shut off feature to automatically shut off the gas in the event of a downstream pipe rupture.

Earthquake Valves:

Provide seismic gas shut off valves on gas services to all buildings which are intended to be used for student housing or as a residence. Seismic gas shut off valves are not required on non-residential buildings.

Gas Meters

- ~~Gas meters~~ may be either diaphragm bellows or turbine type as appropriate for the application. Gas meters shall record in cubic feet.

- ask dennis./chad. what spec to use
- So far. Jason using American meters
- eric V was looking for 2" remote head meter

remote read may be an option